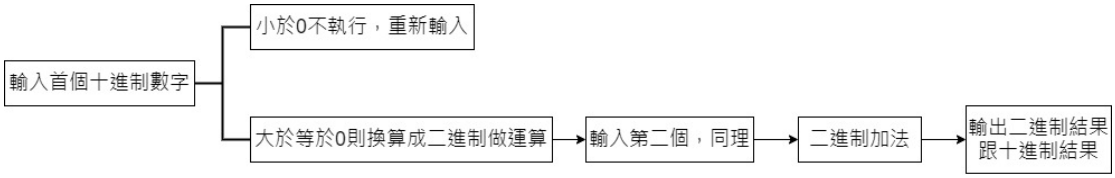
PA2: Positive Binary Adder Review

silver-kite-2a91

1. 此次程式碼大致如此運作: 輸入兩個十進制的數字，然後程式碼會將兩個數字轉為二進制的數字，然後做二進制的相加進位，最後輸出二進制相加跟輸出二進制總和轉回時進制的值。



- 公告中有提到輸出的十進制相加必須是由二進制轉換而成，所以我最後在輸出十進制的時候有將二進制保留，寫一個二進制 to 十進制。

```
65      sum = dectobin(decimal1);
66      cout << "Calculating... " << sum << " + " << dectobin(decimal2)
67      sum = binaryadd(sum, dectobin(decimal2));
68      cout << "Partial Sum: " << bintodec(sum) << endl << endl;
```

```
178 long long bintodec (long long binsum)
179 {
180     long long decimal = 0;
181     long long n = 1;
182     while(binsum > 0)
183     {
184         decimal += (binsum%10)*n;
185         binsum = binsum/10;
186         n *= 2;
187     }
188     return decimal;
```

上圖第 68 行即為將二進制的總和轉為十進制(binary to decimal(sum))
下圖為二進制回十進制的函數

- 每一次輸入前我都有設置輸入-1 後離開終端機的機制
- ❁❁❁

- 程式結構

```
1    #include <iostream>
2    #include <limits>
3    using namespace std;
4    long long dectobin (long long);
5    long long bintodec (long long);
6    long long binaryadd (long long, long long);
```

前 6 行:相關函式庫、函數宣告

```
8    int main()
9    {
10       int decimal1 = 0;
11       int decimal2 = 0;
12       int sum = 0;
13       int n = 4;
14
15       cout << "*** Welcome to Positive Binary adder ***" << endl;
```

8~15 行:進入主函數宣告相關變數，輸出”歡迎進入系統”

```
17       //1st number
18       cout << "Please enter the 1st number in decimal: ";
19       cin >> decimal1;
20       while(!cin.good())
21           {
22               cout << "Fool-proof design activated." << endl << "Please enter the 1st decimal number again: ";
23               cin.clear();
24               cin.ignore(numeric_limits<streamsize>::max(), '\n');
25               cin >> decimal1;
26           }
27       if(decimal1 == -1)
28       {
29           cout << "Bye~";
30           return 0;
31       }
32       else{}
33       while(decimal1 < 0 || decimal1 > 524287)
34       {
35           cout << "The decimal number you entered is out of range. Please enter the 1st decimal number again: ";
36           cin >> decimal1;
37       }
```

17~37 行: 輸入第一個數字並檢查範圍

```
39       //2nd number
40       cout << "Please enter the 2nd number in decimal: ";
41       cin >> decimal2;
42       while(!cin.good())
43           {
44               cout << "Fool-proof design activated." << endl << "Please enter the 2nd decimal number again: ";
45               cin.clear();
46               cin.ignore(numeric_limits<streamsize>::max(), '\n');
47               cin >> decimal2;
48           }
49       if(decimal2 == -1)
50       {
51           cout << "Bye~";
52           return 0;
53       }
54       else{}
55       while(decimal2 < 0|| decimal2 > 524287)
56       {
57           cout << "The decimal number you entered is out of range. Please enter the 2nd decimal number again: ";
58           cin >> decimal2;
59       }
60       while(decimal1 + decimal2 > 524287)
61       {
62           cout << "The partial sum is out of range. Please enter the 2nd decimal number again: ";
63           cin >> decimal2;
64       }
65       sum = dectobin(decimal1);
66       cout << "Calculating... " << sum << " + " << dectobin(decimal2) << " = " << binaryadd(sum, dectobin(decimal2)) << endl;
67       sum = binaryadd(sum, dectobin(decimal2));
68       cout << "Partial Sum: " << bintodec(sum) << endl << endl;
```

39~68 行: 輸入第二個數字並檢查範圍，然後將兩個數字加起來

```
70      //3rd number
71      cout << "Please enter the 3rd number in decimal: ";
72      cin >> decimal1;
73      while(!cin.good())
74          {
75              cout << "Fool-proof design activated." << endl << "Please enter the 3rd decimal number again: ";
76              cin.clear();
77              cin.ignore(numeric_limits<streamsize>::max(), '\n');
78              cin >> decimal1;
79          }
80      if(decimal1 == -1)
81      {
82          cout << "Bye~";
83          return 0;
84      }
85      else{}
86      while(decimal1 < 0 || decimal1 > 524287)
87      {
88          cout << "The decimal number you entered is out of range. Please enter the 3rd decimal number again: ";
89          cin >> decimal1;
90      }
91      while(decimal1 + decimal2 > 524287)
92      {
93          cout << "The partial sum is out of range. Please enter the 3rd decimal number again: ";
94          cin >> decimal1;
95      }
96      cout << "Calculating... " << sum << " + " << dectobin(decimal1) << " = " << binaryadd(sum, dectobin(decimal1)) << endl;
97      sum = binaryadd(sum, dectobin(decimal1));
98      cout << "Partial Sum: " << bintodec(sum) << endl << endl;
```

70~98 行：輸入第三個數字並檢查範圍，符合後算出加總

```
101     //4,5,6...th number
102     while(true)
103     {
104         cout << "Please enter the " << n << "th number in decimal: ";
105         cin >> decimal2;
106         while(!cin.good())
107         {
108             cout << "Fool-proof design activated." << endl << "Please enter the " << n << "th decimal number again: ";
109             cin.clear();
110             cin.ignore(numeric_limits<streamsize>::max(), '\n');
111             cin >> decimal2;
112         }
113         if(decimal2 == -1)
114         {
115             cout << "Bye~";
116             return 0;
117         }
118         else{}
119         while(decimal2 < 0 || decimal2 > 524287)
120         {
121             cout << "The decimal number you entered is out of range. Please enter the " << n << "th decimal number again: ";
122             cin >> decimal2;
123         }
124         while(decimal1 + decimal2 > 524287)
125         {
126             cout << "The partial sum is out of range. Please enter the " << n << "th decimal number again: ";
127             cin >> decimal2;
128         }
129         n = n + 1;
130         cout << "Calculating... " << sum << " + " << dectobin(decimal2) << " = " << binaryadd(sum, dectobin(decimal2)) << endl;
131         sum = binaryadd(sum, dectobin(decimal2));
132         cout << "Partial Sum: " << bintodec(sum) << endl << endl;
```

```
133         cout << "Please enter the " << n << "th number in decimal: ";
134         cin >> decimal1;
135         while(!cin.good())
136         {
137             cout << "Fool-proof design activated." << endl << "Please enter the " << n << "th decimal number again: ";
138             cin.clear();
139             cin.ignore(numeric_limits<streamsize>::max(), '\n');
140             cin >> decimal1;
141         }
142         if(decimal1 == -1)
143         {
144             cout << "Bye~";
145             return 0;
146         }
147         else{}
148         while(decimal1 < 0 || decimal1 > 524287)
149         {
150             cout << "The decimal number you entered is out of range. Please enter the " << n << "th decimal number again: ";
151             cin >> decimal1;
152         }
153         while(decimal1 + decimal2 > 524287)
154         {
155             cout << "The partial sum is out of range. Please enter the " << n << "th decimal number again: ";
156             cin >> decimal1;
157         }
158         n = n + 1;
159         cout << "Calculating... " << sum << "+" << dectobin(decimal1) << "=" << binaryadd(sum, dectobin(decimal1)) << endl;
160         sum = binaryadd(sum, dectobin(decimal1));
161         cout << "Partial Sum: " << bintodec(sum) << endl << endl;
162     }
163     return 0;
164 }
```

輸入接下來的數字(第四第五第六…)，使用”n”th 表示

(紅圈)

2. 幾經修改，公告的一筆測資丟入我的程式碼執行通過。
3. 這次的難處在於**二進制相加系統**，一開始構思出來的系統總是卡在某個地方，後來查看網路資料才知道自己錯誤的地方。
4. 額外功能: 防止惡意輸入機制

```
20      while(!cin.good())
21      {
22          cout << "Fool-proof design activated." << endl << "Ple
23          cin.clear();
24          cin.ignore(numeric_limits<streamsize>::max(), '\n');
25          cin >> decimall;
26      }
```

如果你輸入的不是數字(比如字母)，通常 code 都會開始當機，具體原因我也不知道是甚麼，不過我**設計了防止惡意輸入的程式碼，不管輸入甚麼字母符號(cin 會顯示 bad)，不符合會要求重新輸入。**更多有意義的功能待開發...

5. 我認為這次的作業相比於上次作業目標簡單明確，僅此。
6. 由於 codeblock 會吃掉我的註解，所以我將程式碼以及註解一起放在下方供查閱。(主程式、註解、函數)

```
#include <iostream>
using namespace std;
long long dectobin (long long);
long long bintodec (long long);
long long binaryadd (long long, long long);

int main()
{
    long long decimal1 = 0;
    long long decimal2 = 0;
    long long sum = 0;
    int n = 4;

    cout << "*** Welcome to Positive Binary adder ***" << endl;

    //1st number
    cout << "Please enter the 1st number in decimal: ";
    cin >> decimal1;
    if (decimal1 == -1)
    {
        cout << "Bye~";
        return 0;
```

```
	}
	while(decimal1 < 0 || decimal1 > 524287)
	{
		cout << "The decimal number you entered is out of range. Please enter the 1st decimal number again: ";
		cin >> decimal1;
	}

	//2nd number
	cout << "Please enter the 2nd number in decimal: ";
	cin >> decimal2;
	if (decimal2 == -1)
	{
		cout << "Bye~";
		return 0;
	}
	while(decimal2 < 0 || decimal2 > 524287)
	{
		cout << "The decimal number you entered is out of range. Please enter the 2nd decimal number again: ";
		cin >> decimal2;
	}
	while(decimal1 + decimal2 > 524287)
	{
		cout << "The partial sum is out of range. Please enter the 2nd decimal number again: ";
		cin >> decimal2;
	}
	sum = dectobin(decimal1);
	cout << "Calculating... " << sum << " + " << dectobin(decimal2) << " = " << binaryadd(sum, dectobin(decimal2)) << endl;
	sum = binaryadd(sum, dectobin(decimal2));
	cout << "Partial Sum: " << bintodec(sum) << endl << endl;

	//3rd number
	cout << "Please enter the 3rd number in decimal: ";
	cin >> decimal1;
	if (decimal1 == -1)
```

```
	{
		cout << "Bye~";
		return 0;
	}
	while(decimal1 < 0 || decimal1 > 524287)
	{
		cout << "The decimal number you entered is out of range. Please enter the 3rd decimal number again: ";
		cin >> decimal1;
	}
	while(decimal1 + decimal2 > 524287)
	{
		cout << "The partial sum is out of range. Please enter the 3rd decimal number again: ";
		cin >> decimal1;
	}
	cout << "Calculating... " << sum << " + " << dectobin(decimal1) << " = " << binaryadd(sum, dectobin(decimal1)) << endl;
	sum = binaryadd(sum, dectobin(decimal1));
	cout << "Partial Sum: " << bintodec(sum) << endl << endl;


	//4,5,6...th number
	while(true)
	{
		cout << "Please enter the " << n << "th number in decimal: ";
		cin >> decimal2;
		if (decimal2 == -1)
		{
			cout << "Bye~";
			break;
		}
		while(decimal2 < 0 || decimal2 > 524287)
		{
			cout << "The decimal number you entered is out of range. Please enter the " << n << "th decimal number again: ";
			cin >> decimal2;
		}
```

```
            while(decimal1 + decimal2 > 524287)
            {
                cout << "The partial sum is out of range. Please enter the " << n
<< "th decimal number again: ";
                cin >> decimal2;
            }
            n = n + 1;
            cout << "Calculating... " << sum << " + " << dectobin(decimal2) << " = "
<< binaryadd(sum, dectobin(decimal2)) << endl;
            sum = binaryadd(sum, dectobin(decimal2));
            cout << "Partial Sum: " << bintodec(sum) << endl << endl;
            cout << "Please enter the " << n << "th number in decimal: ";
            cin >> decimal1;
            if (decimal1 == -1)
            {
                cout << "Bye~";
                break;
            }
            while(decimal1 < 0 || decimal1 > 524287)
            {
                cout << "The decimal number you entered is out of range. Please
enter the 3rd decimal number again: ";
                cin >> decimal1;
            }
            while(decimal1 + decimal2 > 524287)
            {
                cout << "The partial sum is out of range. Please enter the " << n
<< "th decimal number again: ";
                cin >> decimal1;
            }
            n = n + 1;
            cout << "Calculating... " << sum << "+" << dectobin(decimal1) << "="
<< binaryadd(sum, dectobin(decimal1)) << endl;
            sum = binaryadd(sum, dectobin(decimal1));
            cout << "Partial Sum: " << bintodec(sum) << endl << endl;
        }
        return 0;
}
```

```
long long dectobin (long long number)//decimal to binary
{
	long long binary = 0;
	long long i = 1;
	while(number > 0)
	{
		binary += (number%2)*i;
		number = number/2;
		i *= 10;
	}
	return binary;
}
long long bintodec (long long binsum)
{
	long long decimal = 0;
	long long n = 1;
	while(binsum > 0)
	{
		decimal += (binsum%10)*n;
		binsum = binsum/10;
		n *= 2;
	}
	return decimal;
}
long long binaryadd (long long a1, long long a2)// addition of two binary
numbers
{
	long long x = max(a1,a2);
	long long y = min(a1,a2);
	long long result = 0;
	int carry = 0;
	long long place = 1;
	while(x >= 0)
	{
		a1 = x%10;
		a2 = y%10;
		result = ((a1+a2+carry)%2)*place + result;
```

```
            carry = (a1+a2+carry)/2;
            x = x/10;
            y = y/10;
            place = place*10;
            if(x == 0 && carry == 0)
                  break;
      }
      return result;
}
```